

A "fact" about fractions

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I was checking my work on a problem recently^[1], where I reached the equation

$$-\frac{1}{8} - \frac{3}{4} - \frac{3}{2} + \frac{1}{24} \stackrel{?}{=} -\frac{7}{3}.$$

At this point I concluded that I must have made a mistake. Without even calculating, I knew the left side had denominator 24, but the right side had denominator 3.

...Or so I thought. Actually, the two sides are equal.

Is it intuitive that the denominator, in lowest terms, can decrease? Maybe—the example

$$\frac{1}{3} + \frac{1}{6} = \frac{1}{2}$$

is fairly common.

But I had to ask: why did I make this wrong assumption? Is there some other true statement we can conclude?

An Investigation

Question.

Are there integers a , b , c that make the following equation true?

$$8a + 4b + 2c + 1 = 56$$

Clearly, the answer is no: taking both sides modulo 2 gives $0 \equiv 1 \pmod{2}$, which is always false. This idea, wrongly applied to the fractional example, caused me to dismiss a correct solution.

But, returning to the original problem, there is something we can say:

① **Valid Denominators.**

Find all possible denominators, in simplest form, of

$$\frac{a}{8} + \frac{b}{4} + \frac{c}{2} + \frac{d}{24}$$

where $a, b, c, d \in \mathbb{Z}$.

The answer is **all divisors of 24**. Notice that

$$\frac{a}{8} + \frac{b}{4} + \frac{c}{2} + \frac{d}{24} = \frac{3a + 6b + 12c + d}{24},$$

and the numerator can be any integer. So, the possible denominators are those that divide 24.

In general, given an expression $\frac{p_1}{q_1} + \frac{p_2}{q_2} + \dots + \frac{p_n}{q_n}$, all we can say is that the denominator in lowest terms is a divisor of $\text{lcm}(q_1, \dots, q_n)$.

This isn't a very strong statement, but at least it's something. For example, we can quickly show that the following equation has no solutions for $a, b \in \mathbb{Z}$.

$$\frac{a}{8} + \frac{b}{3} = \frac{1}{48}$$

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1. The problem was finding the partial fraction decomposition of $\frac{1+x+x^2}{(1-x)^3(1+x)}$. I found the correct answer and plugged in $x = 2$ to check, leading to the main equation. ↩